

Nine-Valent Human Papillomavirus Vaccination and Related Cancers in Males

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 Supplemental content

IMPORTANCE The disease burden of human papillomavirus (HPV)-related cancers is substantial, irrespective of sex. Establishing evidence on the effectiveness of the 9-valent HPV vaccine for males may support the implementation of sex-neutral HPV vaccination strategies.

OBJECTIVE To evaluate the incidence of HPV-related cancers between 9-valent HPV-vaccinated and unvaccinated males using a large-scale database.

DESIGN, SETTING, AND PARTICIPANTS This multicenter retrospective cohort study utilized a global database. Participants included males aged 9 to 26 years who were either unvaccinated or received at least one 9-valent HPV vaccine between January 2016 and December 2024 with outcomes followed-up for up to 10 years.

MAIN OUTCOMES AND MEASURES The primary composite outcomes were HPV-related cancers, including head and neck, esophageal, anal, and penile cancers. Hazard ratios (HRs) were estimated using propensity score matching.

RESULTS Before propensity score matching, 615 155 vaccinated males (mean [SD] age, 13.4 [3.5] years) and 2 290 623 unvaccinated males (mean [SD] age, 17.2 [5.5] years) were identified in the US. After propensity score matching, 510 260 participants were included in each group. Compared with those in the unvaccinated group, those in the vaccinated group had a lower risk of the primary composite outcome (HR, 0.54; 95% CI, 0.37-0.81; $P = .002$). Subgroup analysis demonstrated that the significantly lower incidence of the primary composite outcome in the vaccinated group was maintained among both males aged 9 to 14 years (HR, 0.58; 95% CI, 0.34-0.97; $P = .04$) and those aged 15 to 26 years (HR, 0.50; 95% CI, 0.27-0.93; $P = .03$).

CONCLUSIONS AND RELEVANCE This study found a significant negative association of the 9-valent HPV vaccine with HPV-related cancers in adolescents and young adult males and contributes to the development of sex-neutral HPV vaccination.

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Human papillomavirus (HPV) is an oncogenic virus associated with substantial morbidity and mortality worldwide.^{1,2} HPV infection is a major cause of genital cancers (including cervical, vaginal, vulvar, penile, and anal cancers), as well as head and neck cancers (such as oropharyngeal cancer).^{1,3} The disease burden of HPV infection is enormous for both females and males.⁴

Vaccination is one of the most effective strategies to prevent HPV infection and related diseases.⁵ Some studies have reported that vaccination is highly effective in preventing HPV infection, precancerous cervical lesions, and cancers among females.⁶⁻⁹ Although the preventive impact of HPV vaccination against HPV-related cancers has been well documented in females, evidence in males remains limited. Most published studies involving males have assessed the impact of HPV vaccination on HPV infection or the overall incidence

of HPV-related cancers before and after the introduction of the HPV vaccine.⁹⁻¹² Evidence of the impact of male HPV vaccination on cancer by comparing vaccinated and unvaccinated cohorts with a large sample size has not been well established. Some countries include HPV vaccination in their national immunization programs for both males and females.¹³ However, sex-neutral HPV vaccination has not yet been implemented in many other countries. Establishing evidence regarding HPV vaccine effectiveness for males may facilitate sex-neutral HPV vaccination, leading to further reduction of the HPV-related disease burden worldwide.

Initially, 2- and 4-valent HPV vaccines were available, followed by the approval of a 9-valent HPV (9v-HPV) vaccine.¹⁴ The 9v-HPV vaccine has accounted for the majority of the HPV vaccination market since 2016 in several countries, including the US, Europe, and Australia.^{15,16} Given that the 9v-HPV vac-

cine is expected to become a mainstay of HPV vaccination, robust evidence regarding its effectiveness in HPV-related cancers is needed. The objective of this study was to evaluate the incidence of HPV-related cancers in 9v-HPV-vaccinated and unvaccinated males using a large-scale database.

Methods

Study Design and Settings

This retrospective cohort study followed the Strengthening of Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline. We used the TriNetX platform, a large-scale global database, to evaluate the association of 9v-HPV vaccination for adolescent and young males aged 9 to 26 years with HPV-related cancers, including head and neck, penile, rectal, and anal cancers. TriNetX has stored electronic health records for approximately 190 million patients from 170 health care centers globally as of December 2025.¹⁷ Data collected in TriNetX included health care visits, diagnoses, procedures, medications, and laboratory results. In this study, the vaccinated cohort included males aged 9 to 26 years receiving at least one 9v-HPV vaccination with a health examination between January 2016 and December 2024. The unvaccinated cohort included males aged 9 to 26 years who underwent a health examination during the corresponding period but never received any HPV vaccination. Age categorization was based on the current recommendation for HPV vaccination.^{18,19} Details of the data extraction code and cohort selection flowchart are provided in eTable 1 and eFigure 1 in Supplement 1.

The primary outcome in this study was a composite measure encompassing any diagnosis of the following cancers, as defined by the *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10)*: head and neck cancers, penile cancer, esophageal cancer (upper two-thirds only), and anal cancer (eTable 2 in Supplement 1).²⁰ The secondary outcomes included the individual diagnoses of each of the 4 cancers. Data were extracted on February 14, 2026. Outcome follow-up started 180 days after the index date. The index date was defined as the date of the first dose with health examination in the vaccinated group and the date of the first health examination in the unvaccinated group.

The TriNetX platform complied with the Health Insurance Portability and Accountability Act and General Data Protection Regulation.²¹ Therefore, informed consent was not required because this study utilized only deidentified and aggregated patient data. Although deidentified TriNetX studies generally do not require institutional review board approval,²² this study was reviewed and approved for publication by the institutional review board of Nara Prefecture General Medical Center.

Statistical Analysis

Data were analyzed by region (ie, the US; Asia and the Pacific; Europe, the Middle East and Africa; and Latin America) as defined by the TriNetX platform. Standardized mean differ-

Key Points

Question Is the 9-valent human papillomavirus (9v-HPV) vaccine associated with lower risk of cancer in males?

Findings This cohort study of 615 155 vaccinated and 2 290 623 unvaccinated males aged 9 to 26 years found that males vaccinated with the 9v-HPV vaccine had a lower risk of HPV-related cancer compared with the unvaccinated cohort. The negative association of the 9v-HPV vaccine with HPV-related cancer was maintained in both males aged 9 to 14 years and males aged 15 to 26 years.

Meaning The findings of this study suggest that the 9v-HPV vaccine is beneficial not only for females but also for males.

ences were estimated to evaluate the balance of each characteristic between the 2 groups. Propensity score matching (PSM) using the age group, race, and underlying medical conditions as covariates. Age group was matched by 2-year bands. In the TriNetX platform, race was categorized as American Indian or Alaska Native, Asian, Black or African American, Native Hawaiian or Other Pacific Islander, White, other race (any race other than the 5 defined races) or unknown. However, American Indian and Alaska Native and Native Hawaiian and Other Pacific Islander could not be included in PSM in this study due to the limitation in the number of covariates for PSM by the TriNetX built-in platform. Race was included as a demographic variable to account for known disparities. Sex was classified as female, male, or unknown. Race and sex data were reported by health care centers contributing to the TriNetX platform. Underlying medical conditions included in the PSM were family history of malignant neoplasms, nicotine dependence, alcohol related disorders, high-risk sexual behavior, high-risk socioeconomic and psychosocial circumstances, chronic lower respiratory diseases, developmental disorders, diseases of oral cavity and salivary glands, esophagitis, diseases of liver, obesity, diabetes, neurological diseases, diseases of genital organs, kidney diseases, neoplasms, immunocompromising conditions, sexually transmitted infections, and HIV disease, which were selected based on previous studies that evaluated high-risk factors for HPV-related cancers.²³⁻²⁶ The underlying medical conditions were extracted based on *ICD-10* coding. The ascertainment window for underlying medical conditions was defined as until the day of the index date. PSM was implemented using the greedy nearest-neighbor pairing method with a 1:1 ratio and a pooled SD caliper of 0.1. The hazard ratio (HR) with a 95% CI for the vaccinated group compared with the unvaccinated group was then calculated. A Cox proportional hazards model was applied in the analysis using R version 4.0.2 (R Project for Statistical Computing) with the survival package 3.2-3 (eMethods in Supplement 1). The log-rank test was used to compare the survival curves of outcome-free participants between the vaccinated and unvaccinated groups. The cutoff for statistical significance was a 2-sided $P < .05$ in this study.

Subgroup analyses were conducted by age group (ie, 9-14 and 15-26 years at the time of the index date) with age group

Table 1. Characteristics of Vaccinated and Unvaccinated Participants Before Propensity Score Matching

Characteristic	Participants, No. (%)		SMD
	Vaccinated (n = 615 155)	Unvaccinated (n = 2 290 623)	
Demographics			
Age, y			
Mean (SD)	13.4 (3.5)	17.2 (5.5)	0.830
9-10	47 283 (7.7)	387 465 (16.9)	0.284
11-12	296 360 (48.2)	200 263 (8.7)	0.972
13-14	110 171 (17.9)	216 601 (9.5)	0.248
15-16	64 042 (10.4)	249 222 (10.9)	0.015
17-18	43 606 (7.1)	257 872 (11.3)	0.145
19-20	17 158 (2.8)	224 508 (9.8)	0.292
21-22	13 121 (2.1)	239 387 (10.5)	0.348
23-24	11 551 (1.9)	250 973 (11.0)	0.377
25-26	11 863 (1.9)	264 332 (11.5)	0.391
Race			
Asian	29 437 (4.8)	90 754 (4.0)	0.040
Black or African American	124 153 (20.2)	378 508 (16.5)	0.095
White	338 584 (55.0)	1 332 808 (58.2)	0.063
Other race ^a	52 114 (8.5)	149 881 (6.5)	0.073
Unknown	63 540 (10.3)	312 543 (13.6)	0.102
Medical conditions ^b			
Family history of primary malignant neoplasm (Z80)	3611 (0.6)	13 071 (0.6)	0.002
Nicotine dependence (F17)	4568 (0.7)	73 726 (3.2)	0.178
Alcohol related disorders (F10)	1889 (0.3)	28 462 (1.2)	0.107
High-risk sexual behavior (Z72.5)	3320 (0.5)	12 830 (0.6)	0.003
High-risk socioeconomic and psychosocial circumstances (Z55-Z65)	46 625 (7.6)	76 482 (3.3)	0.187
Chronic lower respiratory diseases (J40-J4A)	135 124 (22.0)	267 012 (11.7)	0.278
Developmental disorders (F80-F89)	77 612 (12.6)	154 704 (6.8)	0.199
Diseases of oral cavity and salivary glands (K00-K14)	65 977 (10.7)	134 720 (5.9)	0.176
Esophagitis (K20)	2349 (0.4)	10 653 (0.5)	0.013
Diseases of liver (K70-K77)	3436 (0.6)	17 842 (0.8)	0.027
Obesity (E66)	93 488 (15.2)	160 212 (7.0)	0.263
Diabetes mellitus (E08-E13)	3857 (0.6)	23 983 (1.0)	0.046
Neurological diseases (G00-G99)	107 842 (17.5)	321 236 (14.0)	0.096
Diseases of male genital organs (N40-N53)	44 443 (7.2)	111 658 (4.9)	0.099
Acute kidney failure and chronic kidney disease (N17-N19)	2617 (0.4)	19 407 (0.8)	0.053
Neoplasms (C00-D49)	26 832 (4.4)	88 237 (3.9)	0.026
Certain disorders involving the immune mechanism (D80-D89)	3021 (0.5)	13 560 (0.6)	0.014
Sexually transmitted infections (A50-A64)	2964 (0.5)	15 842 (0.7)	0.027
HIV disease (B20)	1352 (0.2)	3587 (0.2)	0.015

Abbreviation: SMD, standardized mean difference.

^a American Indian or Alaska Native, Native Hawaiian or other Pacific Islander, or any other race could not be included in covariates of propensity score matching due to the limitation in the TriNetX built-in platform regarding the number of covariates in the matching.

^b *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* codes for each medical condition are listed in parentheses.

matched by single-year bands. Sensitivity analyses were performed by modifying the initial follow-up period to 1 year and 2 years after the index date. Additional supplemental analyses were performed by replacing 9v-HPV vaccines with 2- or 4-valent HPV vaccines to investigate the association of 2- and 4-valent HPV vaccines with HPV-related cancers. Two- and 4-valent vaccines were administered before 2016. The index period for the 2- and 4-valent HPV vaccine was between January 2012 and December 2015. The statistical analyses described were conducted using the built-in analytical tools of the TriNetX platform.

Results

In the US, 615 155 vaccinated males (mean [SD] age, 13.4 [3.5] years) and 2 290 623 unvaccinated males (mean [SD] age, 17.2 [5.5] years) aged 9 to 26 years were identified. The participant characteristics before PSM are presented in **Table 1**. In the vaccinated and unvaccinated groups in the US, 296 360 (48.2%) and 200 263 (8.7%) were aged 11 to 12 years, respectively. For race, 29 437 (4.8%) and 90 754 (4.0%) were identified as Asian, 124 153 (20.2%) and 378 508 (16.5%) as Black or

Table 2. Characteristics of Vaccinated and Unvaccinated Participants After Propensity Score Matching

Characteristic	Participants, No. (%)		SMD
	Vaccinated (n = 510 260)	Unvaccinated (n = 510 260)	
Demographics			
Age, y			
Mean (SD)	13.8 (3.7)	13.9 (3.7)	0.020
9-10	47 274 (9.3)	47 204 (9.3)	<0.001
11-12	194 331 (38.1)	191 167 (37.5)	0.013
13-14	107 521 (21.1)	109 538 (21.5)	0.010
15-16	63 886 (12.5)	64 831 (12.7)	0.006
17-18	43 575 (8.5)	44 005 (8.6)	0.003
19-20	17 153 (3.4)	17 209 (3.4)	0.001
21-22	13 118 (2.6)	13 063 (2.6)	0.001
23-24	11 546 (2.3)	11 513 (2.3)	<0.001
25-26	11 856 (2.3)	11 730 (2.3)	0.002
Race			
Asian	21 565 (4.2)	21 900 (4.3)	0.003
Black or African American	93 469 (18.3)	93 133 (18.3)	0.002
White	293 552 (57.5)	295 703 (58.0)	0.009
Other race ^a	37 700 (7.4)	37 709 (7.4)	<0.001
Unknown	57 949 (11.4)	56 221 (11.0)	0.011
Medical conditions ^b			
Family history of primary malignant neoplasm (Z80)	2800 (0.5)	2955 (0.6)	0.004
Nicotine dependence (F17)	4495 (0.9)	4516 (0.9)	<0.001
Alcohol related disorders (F10)	1865 (0.4)	1874 (0.4)	<0.001
High-risk sexual behavior (Z72.5)	3162 (0.6)	3356 (0.7)	0.005
High-risk socioeconomic and psychosocial circumstances (Z55-Z65)	28 054 (5.5)	28 819 (5.6)	0.007
Chronic lower respiratory diseases (J40-J4A)	89 215 (17.5)	91 376 (17.9)	0.011
Developmental disorders (F80-F89)	52 457 (10.3)	52 889 (10.4)	0.003
Diseases of oral cavity and salivary glands (K00-K14)	41 899 (8.2)	41 518 (8.1)	0.003
Esophagitis (K20)	2083 (0.4)	2203 (0.4)	0.004
Diseases of liver (K70-K77)	2852 (0.6)	2991 (0.6)	0.004
Obesity (E66)	57 092 (11.2)	58 239 (11.4)	0.007
Diabetes mellitus (E08-E13)	3577 (0.7)	3748 (0.7)	0.004
Neurological diseases (G00-G99)	80 620 (15.8)	84 778 (16.6)	0.022
Diseases of male genital organs (N40-N53)	29 959 (5.9)	30 477 (6.0)	0.004
Acute kidney failure and chronic kidney disease (N17-N19)	2390 (0.5)	2357 (0.5)	0.001
Neoplasms (C00-D49)	21 204 (4.2)	22 059 (4.3)	0.008
Certain disorders involving the immune mechanism (D80-D89)	2680 (0.5)	2696 (0.5)	<0.001
Sexually transmitted infections (A50-A64)	2763 (0.5)	2822 (0.6)	0.002
HIV disease (B20)	1242 (0.2)	1256 (0.2)	0.001

Abbreviation: SMD, standardized mean difference.

^a American Indian or Alaska Native, Native Hawaiian or other Pacific Islander, or any other race could not be included in covariates of propensity score matching due to the limitation in the TriNetX built-in platform regarding the number of covariates in the matching.

^b *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* codes for each medical condition are listed in parentheses.

African American, 338 584 (55.0%) and 1 332 808 (58.2%) as White, 52 114 (8.5%) and 149 881 (6.5%) as other race, and 63 540 (10.3%) and 312 543 (13.6%) as unknown race in the vaccinated and unvaccinated groups, respectively. The number of vaccinated participants by region is displayed in eTable 3 in Supplement 1. Given the low number of vaccinated males outside the US, subsequent analyses were restricted to the US cohort.

Following PSM, 510 260 participants were included in each group. After PSM, the covariates examined in this study were well-balanced (standardized mean differences <0.1) between the vaccinated and unvaccinated groups (Table 2). After PSM, the primary composite outcome (head and neck cancers, penile

cancer, esophageal cancer, or anal cancer) occurred in 40 and 64 participants in the vaccinated and unvaccinated groups, respectively. The cumulative incidence curves for the primary composite outcome are presented in Figure 1 and eFigure 2 and eFigure 3 in Supplement 1. Compared with the unvaccinated group, the vaccinated group had a lower risk of the primary composite outcome (HR, 0.54; 95% CI, 0.37-0.81; $P = .002$) (Table 3). The results for esophageal, anal, and penile cancers could not be evaluated due to the lack of outcome event.

Subgroup analysis demonstrated a significantly lower incidence of the primary composite outcome in the vaccinated group among both males aged 9 to 14 years (HR, 0.58; 95% CI, 0.34-0.97; $P = .04$) and those aged 15 to 26 years (HR, 0.50;

95% CI, 0.27-0.93; $P = .03$) (Figure 2). In the sensitivity analysis, modifying the follow-up period for outcome starting from 1 year and 2 years after the index date maintained the statistical significance (1 year: HR, 0.52; 95% CI, 0.34-0.80; $P = .002$; 2 years: HR, 0.56; 95% CI 0.35-0.89; $P = .01$). In addition, 2- or 4-valent HPV vaccination was significantly associated with fewer primary outcome events compared with the unvaccinated group (HR, 0.57; 95% CI, 0.35-0.94; $P = .03$).

Discussion

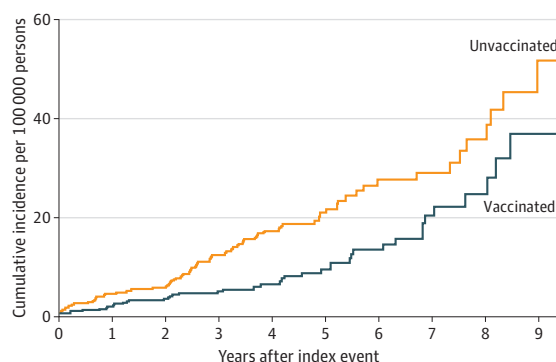
Our cohort study found a significant association of the 9v-HPV vaccine with a reduction in HPV-related cancers in adolescents and young adult males. The statistical significance was maintained both in those aged 9 to 14 years and 15 to 26 years. This finding highlights the benefits of receiving the 9v-HPV vaccine in adolescent and young adult males.

Most studies evaluating the association of HPV vaccination with HPV-related cancers in males have primarily focused on adult populations, who may have already been exposed to HPV, potentially reducing the observed effect.^{9,10,27,28} These studies often assessed 2- or 4-valent HPV vaccines, and/or included relatively small sample sizes (eg, a few thousand or fewer vaccinated participants).²⁷ Given the current recommendations in developed countries advocate vaccinating all adolescent boys with 9v-HPV, our study contributes to the growing body of evidence supporting these current practices.^{13,15} Considering the overall incidence of HPV-related cancers in males irrespective of risk factors, the statistical evaluation of the association of HPV vaccination with HPV-related cancers requires a large sample size. One of the strengths of our study was the large sample size (>600 000 vaccinated males), enabled by the use of the large-scale database.

High-risk sexual behaviors are known to be an established risk factor for developing HPV-related cancers among males.²⁸ Some previous studies have reported that high-risk sexual behaviors are common among males.^{29,30} However, in our study, ICD-10 coding identified that less than 1% of participants had a code associated with high-risk sexual behaviors. Other studies have demonstrated that individuals engaging in high-risk sexual behaviors, including males who have sex with males, were more likely to receive HPV vaccination than those without.³¹ Some guidelines provide a stronger recommendation for males with high-risk sexual behavior to receive catch-up doses of HPV vaccination.^{32,33} These factors could not be completely accounted in our study, leaving the possibility of insufficient adjustment for confounding variables. Future research should include data analyses using databases that can identify high-risk sexual behaviors among adolescents and young adult males. Moreover, given that few events of esophageal, anal, or penile cancer were documented in our study population, further studies are also warranted.

This study has some important implications. Currently, many countries still implement HPV vaccination only for females.^{13,15} Our study also demonstrated that males accounted for approximately 10% to 15% of the vaccinated cohort outside the US. However, sex-neutral vaccination is criti-

Figure 1. Cumulative Incidence Plot of Human Papillomavirus (HPV)-Related Cancer Among Males Aged 9 to 26 Years



Cumulative incidence is presented by 9-valent HPV vaccination status.

cal to minimize the disease burden of HPV infection.³⁴ Our study provides evidence that HPV vaccination should be accessible irrespective of sex. This could also be important to maximize population-level benefit with a limited number of vaccine doses.^{35,36} Because the incidence of HPV-related cancers is highest in older adults, evidence regarding the association of HPV vaccination with cancer should be collected over a longer period (eg, several decades).

Limitations

Our study had certain limitations. First, extracting medical conditions and outcome events based solely on ICD-10 codes was necessary. The diagnosis of cancer relied only on ICD-10 coding, and the diagnosis could not be confirmed with other information, such as histologic grade; this can result in underascertainment and misascertainment. For instance, a study indicated that relying on ICD-10 coding alone for identification of head and neck cancer had sensitivity, specificity and positive predictive value of 68%, 88%, and 62%, respectively.³⁷ In addition, although the TriNetX database had a low rate of missing data for participants' demographics (eg, age, sex, and race), the platform could not differentiate negative status from unknown status for medical comorbidities and outcomes. As a reference, the reported annual incidence rate of head and neck cancer in those aged 15 to 39 years were approximately 2 per 100 000 person-years.³⁸ However, this reported incidence may not be directly compared with our cumulative incidence because participants were censored at their last visit in our study, which elevated the observed incidence. This is an important limitation due to the nature of multicenter electronic health records in TriNetX. Second, our cohorts consisted of those with a health examination. The percentage of adolescents who received a well-child health check in the past year was approximately 80% in the US.³⁹ However, the high prevalence for health examination visits does not completely eliminate the possibility of selection bias. For example, high-risk sexual behaviors cannot be optimally extracted, resulting in a suboptimal adjustment for this risk factor. Third, we could not obtain vaccination records administered outside the TriNetX-participating centers. Therefore, some participants who were classified in the unvaccinated group may have received the HPV

Table 3. Outcome Comparison After Propensity Score Matching Among Males Aged 9 to 26 Years

Outcome ^a	No. (No. per 100 000)		ARR per 100 000 (95% CI) ^b	HR (95% CI)	P value ^c
	Vaccinated group (n = 510 196)	Unvaccinated group (n = 510 036)			
Composite outcome	40 (7.8)	64 (12.5)	4.71 (0.92-7.27)	0.54 (0.37-0.81)	.002
Head and neck cancer	40 (7.8)	64 (12.5)	4.71 (0.92-7.27)	0.54 (0.37-0.81)	.002

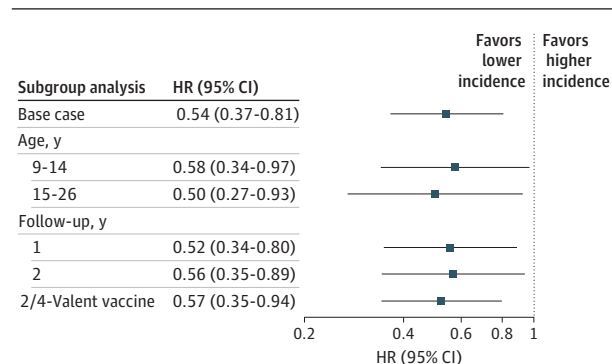
Abbreviations: ARR, absolute risk reduction; HR, hazard ratio; NA, not assessed.

^a Esophageal, anorectal, and penile cancers were additional outcomes but could not be evaluated due to the lack of outcome event.^b Positive values of ARR indicate fewer outcome events in the vaccinated group.

Outcomes with 10 or fewer outcome events in either group could not be assessed.

^c Calculated based on log-rank test to compare cumulative incidence curves between the 2 groups.

Figure 2. Dot Plot of Human Papillomavirus (HPV)-Related Cancer Among Vaccinated Males Compared With Unvaccinated Males



Hazard ratios (HRs) are presented with 95% CIs. Base case analysis is for the HR of 9-valent HPV vaccine among males aged 9 to 26 years. Follow-up since 1 year and 2 years were sensitivity analyses initiating follow-up period from 1 year and 2 years since the index date, respectively.

vaccination. This could explain why the vaccination rate within the study participants was lower than that in the US.^{40,41} In addition, we were unable to obtain information on the number of doses. Currently, a 2-dose schedule is usually recommended for those aged 14 years or younger in many countries, whereas a 3-dose schedule is still implemented for the older population.³² Fourth, our study did not have the data regarding HPV infection status. Not all cancers defined in this study were caused by HPV infection. Systematic reviews indicate that the prevalence of HPV infection in overall head and neck squamous cell carcinomas is approximately 25% (eg, the HPV prevalence is 16% for oral cavity cancer and 14% for laryngeal cancer with much higher

prevalence for oropharyngeal cancer in North America).^{42,43} The HPV-attributable fraction for these cancers could be lower.⁴⁴ In addition, some US studies estimate that the incidence of HPV-related oropharyngeal cancer has continued to increase in older males (45 years or older), whereas that in young males (20-44 years) has begun to decrease; HPV vaccination in young populations and herd immunity for unvaccinated populations are considered to contribute to these changes.^{10,45} These factors made it difficult to assess the direct association of HPV vaccination with only cancers caused by HPV infection. Fifth, the TriNetX platform did not allow us to modify their built-in statistical methodology. For example, the Cox proportional hazard model applied in this study could not eliminate the potential impact of correlation or the matched structure within PSM. A stratified model or cluster-robust variance estimation could have addressed the issue. Sixth, we could not evaluate the data outside the US, given the small number of participants. Therefore, the generalizability of our results outside the US remains limited. Given these limitations the study results need to be interpreted with caution.

Conclusions

In conclusion, this cohort study found a significant negative association of the 9v-HPV vaccine with HPV-related cancers in adolescents and young males. Our study indicates that 9v-HPV vaccines should be administered not only to females but also to young males. This study also contributes to sex-neutral HPV vaccinations. Further studies are needed for long-term follow-up as well as catch-up vaccination in older adults.

ARTICLE INFORMATION

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Author Contributions: Dr Kitano had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Kitano.

Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: Kitano.

Critical review of the manuscript for important intellectual content: All authors.

Statistical analysis: Kitano.

Administrative, technical, or material support:

All authors.

Supervision: Yoshida.

Conflict of Interest Disclosures: Dr Kitano reported receiving grants from BioMérieux outside the submitted work. Dr Yoshida reported receiving grants from TAUNS Laboratories and Mizuho Medy outside the submitted work.

Data Sharing Statement: See Supplement 2.

Additional Contributions: We thank TriNetX for technical assistance supporting cohort building and data extraction.

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